

MealDash

How I Approached the Delivery Time Analysis

Before I wrote a single query

I started with the BRD, not the data. The core business problem was: MealDash's ops team doesn't know why delivery times vary so much, and right now investigating a slow week takes 1-2 days of manual digging. So before writing SQL, I wrote down the 10 questions I needed answered, in the order I thought would build understanding logically start with 'how bad is it overall,' then break it down by the things ops can actually influence (traffic, distance, staffing) versus the things they can't (weather), and end with whether it's getting better or worse over time.

I also decided upfront: I'd report sample sizes next to every average. A 49-minute average from 3 orders means nothing, and I didn't want to make a recommendation I couldn't defend if someone asked 'how many orders is that based on?'

Q1. What is the average and typical (median) delivery time overall?

```
SELECT
  AVG([Time_taken _min_]) AS Avg_Delivery_Time,
  MIN([Time_taken _min_]) AS Min_Delivery_Time,
  MAX([Time_taken _min_]) AS Max_Delivery_Time
FROM MealDash_Clean;
```

First thing I always do is get a baseline before slicing anything. Average came out to 26 minutes, but the range 10 minutes on the fast end, 54 on the slow end told me immediately that averages alone would hide a lot. A 44-minute spread means some segment of orders is having a genuinely bad experience, and the rest of my queries were really about finding which segment that is.

Q2. Which city has the slowest average delivery time?

```
SELECT
  City,
  AVG([Time_taken _min_]) AS Avg_Delivery_Time,
  COUNT(*) AS Total_Orders
FROM MealDash_Clean
GROUP BY City
ORDER BY Avg_Delivery_Time DESC;
```

Semi-Urban came back as the slowest city at 49 minutes nearly double the overall average. My first instinct was that's a huge red flag. But I added COUNT(*) into the query specifically because I've learned not to trust a GROUP BY average without checking how many rows are behind it, and Semi-Urban only had 164 orders out of 45,000+. That's not a trend, that's noise. I made a conscious call not to lead with that number in my recommendations, and instead focused on Metropolitan 27 minutes average, but 34,000+ orders, so a 2-minute improvement there matters far more in absolute terms than 'fixing' Semi-Urban.

Q3. Does road traffic density affect delivery time?

```
SELECT Road_traffic_density, AVG([Time_taken _min_]) AS Avg_Delivery_Time, COUNT(*) AS Total_Orders
FROM MealDash_Clean GROUP BY Road_traffic_density ORDER BY Avg_Delivery_Time DESC;
```

Traffic behaved exactly as expected — Jam at 31 minutes, Low at 21. Nothing surprising, but good to confirm the data isn't lying to me before I trust it for anything else.

Q4. Does weather condition affect delivery time?

```
SELECT Weather_conditions, AVG([Time_taken _min_]) AS Avg_Delivery_Time, COUNT(*) AS Total_Orders
FROM MealDash_Clean GROUP BY Weather_conditions ORDER BY Avg_Delivery_Time DESC;
```

Weather is where I got genuinely surprised. I expected Stormy or Sandstorms to be worst. Instead, Fog and Cloudy topped the list at 28 minutes, while Stormy came in at 25 barely above average. My read on this: agents and ops probably already brace for weather that sounds dramatic, but fog and overcast conditions are the kind of thing nobody plans around, so they cause quiet, unplanned delays. I didn't want to overstate this as fact it's a correlation from aggregate data, not a causal test so I flagged it as something worth a deeper look rather than an immediate action item.

Q5. Does delivery distance affect delivery time?

The dataset only gave me GPS coordinates, not an actual distance. So before I could even ask the question, I had to calculate Distance_km myself using the Haversine formula, which I added as a column back in Power Query. Once that existed, I bucketed distance into ranges and averaged delivery time per bucket.

```
SELECT CASE WHEN Distance_km <= 5 THEN '0-5 km' WHEN Distance_km <= 10 THEN '5-10 km'
      WHEN Distance_km <= 15 THEN '10-15 km' ELSE '15+ km' END AS Distance_Bucket,
      AVG([Time_taken _min_]) AS Avg_Delivery_Time, COUNT(*) AS Total_Orders
FROM MealDash_Clean GROUP BY CASE WHEN Distance_km <= 5 THEN '0-5 km' WHEN Distance_km <= 10
      THEN '5-10 km'
      WHEN Distance_km <= 15 THEN '10-15 km' ELSE '15+ km' END ORDER BY Distance_Bucket;
```

The result wasn't a straight line, which is what I half-expected going in. Orders under 5km average 22 minutes, but once you cross that line, time jumps to the high 20s and then basically flattens 10-15km and 15+km both land around 29 minutes. So it's not that further away = proportionally slower. It's that leaving the immediate zone has a fixed cost, and after that, extra distance barely matters. That changes the recommendation: the fix isn't 'reduce delivery radius everywhere,' it's 'understand what's happening right at that 5km boundary.'

Q6. Do certain vehicle types deliver faster than others?

```
SELECT Type_of_vehicle, AVG([Time_taken _min_]) AS Avg_Delivery_Time, COUNT(*) AS Total_Orders
FROM MealDash_Clean GROUP BY Type_of_vehicle ORDER BY Avg_Delivery_Time DESC;
```

About 3 minutes of difference between fastest and slowest vehicle type, and bicycle only had 68 orders so I didn't weight it heavily. Honestly this query was more about being thorough than expecting a big finding

and that's fine, not every question needs to produce a dramatic result. Ruling something out is still useful information.

Q7. Do higher-rated delivery agents deliver faster?

```
SELECT CASE WHEN Delivery_person_Ratings >= 4.5 THEN '4.5+' WHEN Delivery_person_Ratings >= 4.0 THEN '4.0-4.49'
      WHEN Delivery_person_Ratings >= 3.5 THEN '3.5-3.99' ELSE 'Below 3.5' END AS Rating_Bucket,
      AVG([Time_taken _min_]) AS Avg_Delivery_Time, COUNT(*) AS Total_Orders
FROM MealDash_Clean WHERE Delivery_person_Ratings IS NOT NULL
GROUP BY CASE WHEN Delivery_person_Ratings >= 4.5 THEN '4.5+' WHEN Delivery_person_Ratings >= 4.0 THEN '4.0-4.49'
      WHEN Delivery_person_Ratings >= 3.5 THEN '3.5-3.99' ELSE 'Below 3.5' END ORDER BY
Rating_Bucket DESC;
```

This is the clearest, most consistent pattern in the whole dataset. Agents rated 4.5+ deliver in 24 minutes on average. Every bucket below that sits at 34-37 minutes a 10+ minute gap that doesn't shrink as you go down. I immediately thought about the obvious counter-argument here: maybe agents get rated well BECAUSE they're fast, not the other way around. I can't tell direction of causation from a GROUP BY I'd need to look at the same agent's rating and speed over time to say more. But regardless of which way the arrow points, it's still operationally useful: rating is a strong, easy-to-access signal for who should get time-sensitive orders.

Q8. Does handling multiple deliveries in one trip slow delivery time down?

```
SELECT multiple_deliveries, AVG([Time_taken _min_]) AS Avg_Delivery_Time, COUNT(*) AS
Total_Orders
FROM MealDash_Clean WHERE multiple_deliveries IS NOT NULL GROUP BY multiple_deliveries ORDER
BY multiple_deliveries;
```

0 extra orders: 22 min. 1 extra: 26 min. 2 extra: 40 min. 3 extra: 47 min. What stood out to me is that the jump from 1 to 2 orders (14 minutes) is much bigger than the jump from 0 to 1 (4 minutes). That's not a straight line, it's accelerating each additional stop costs more than the last one, probably because of routing and hand-off overhead stacking up. That distinction mattered for my recommendation: I didn't just say 'reduce bundling,' I said cap it specifically at 1 extra order, because that's where the cost is still small.

Q9. Are festival days slower than normal days?

```
SELECT Festival, AVG([Time_taken _min_]) AS Avg_Delivery_Time, COUNT(*) AS Total_Orders
FROM MealDash_Clean GROUP BY Festival ORDER BY Avg_Delivery_Time DESC;
```

45 minutes on festival days versus 25 on normal days. Almost double. This was the biggest effect size out of all 10 questions, by a wide margin bigger than traffic, weather, distance, or vehicle type combined. Sample size is smaller here (896 orders vs 44,000+), so I wouldn't bet the whole strategy on this single dataset's festival window, but the effect is too large to be noise. This is the one finding I'd put at the top of any recommendation list, because fixing a problem this size, even partially, would move the needle more than any of the smaller factors.

Q10. Is delivery time trending better or worse over time?

```
SELECT Order_Date, AVG([Time_taken _min_]) AS Daily_Avg_Delivery_Time,
       AVG(AVG([Time_taken _min_])) OVER (ORDER BY Order_Date ROWS BETWEEN 6 PRECEDING AND
CURRENT ROW) AS Rolling_7Day_Avg
FROM MealDash_Clean GROUP BY Order_Date ORDER BY Order_Date;
```

Daily averages bounce around 23-30 minutes, which on its own looks alarming if you just eyeball day-to-day numbers. That's exactly why I used a rolling 7-day average instead of reacting to single-day spikes — once smoothed out, the trend sits flat around 25-26 minutes for the whole period. No systemic drift up or down. That's actually a useful conclusion: it means the patterns I found in Q2 through Q9 are structural, not the result of things generally getting worse or better over the time window I had data for.

Pulling it together — what I'd actually tell ops leadership

If I only had five minutes with the ops team, I wouldn't walk through all 10 queries. I'd say this: three things matter more than the rest — festival-day demand, order-bundling limits, and agent-rating-based routing because they're the largest, most actionable, and least ambiguous effects I found. Traffic and weather matter too, but they're conditions ops can't control directly; the first three are things ops CAN act on this week.

Findings summary

Q#	Question	What I found	How confident I am
1	Overall delivery time	26 min avg, 10-54 min range	High — full dataset
2	Slowest city	Metropolitan slowest by volume (27 min)	High for Metro; low for Semi-Urban (164 orders)
3	Traffic impact	Jam adds ~10 min vs Low	High — large, consistent groups
4	Weather impact	Fog/Cloudy slowest, not storms	Medium — surprising, needs follow-up
5	Distance impact	Cost is crossing 5km, not distance itself	High — clear plateau pattern
6	Vehicle type impact	Minor, ~3 min	Medium — bicycle sample too small
7	Agent rating impact	4.5+ agents 10+ min faster	High effect, but direction of cause unclear
8	Bundled orders impact	Non-linear, accelerates after 1 extra	High — large, consistent gap
9	Festival impact	80% slower on festival days	High effect size, moderate sample
10	Trend over time	Flat, no drift	High — smoothed over full period